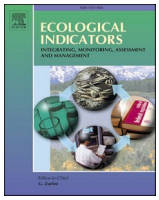
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Experiential factors associated with subjective well-being in urban waterfront environments: Evidence from social media data in Wuhan, China

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ABSTRACT

Urban waterfront environments, as important forms of urban blue spaces, are increasingly recognized for their contributions to human health and subjective well-being (SWB). However, previous studies have mainly classified waterfront spaces based on waterbody attributes and examined the relationships between objectively measured environmental characteristics and health outcomes, while how spatial forms and functional attributes are associated with public experiences and how these experiential differences relate to SWB remain underexplored. To address these gaps, this study utilizes large-scale social media data from Wuhan to explore how experiential factors—including perception, cognition, and behavior—are associated with SWB in urban waterfront environments and examines their temporal dynamics and typological disparities. The results reveal pronounced long-term variations and seasonal cycles in SWB across waterfront environments that are associated with social and climatic conditions. Overall, physical activity shows the strongest association with SWB (19.12%), followed by traffic (15.96%) and crowding (9.89%). Distinct patterns emerge across waterfront typologies: riverfronts, characterized by a balanced mix of exercise and leisure, exhibit the highest overall SWB with moderate seasonality; waterfront parks, more strongly associated with accessibility and natural environmental quality, show slightly lower SWB but the greatest year-round stability; and waterfront greenways, characterized by physical activity, display the lowest levels of SWB with the largest seasonal variability. This study highlights general experiential patterns and typology-specific differences associated with SWB in waterfront environments. It provides empirical evidence for understanding the health-related benefits of urban blue spaces and offers practical insights for the design and management of waterfront environments to promote public well-being.

1. Introduction

Urbanization has generated substantial economic and social benefits, yet it has also intensified environmental pressures and health challenges (Flies et al., 2019; Lorenc et al., 2012). This has spurred growing interest in how urban environments can be planned to support population health and advance sustainable development agendas (Adlakha and John, 2022; Hua et al., 2022). Among various types of urban public spaces, waterfront environments play a particularly important role. In addition to regulating microclimates and improving ecological conditions, they provide valuable settings for recreation, physical activity, and social interaction, thereby demonstrating considerable potential to promote human health and well-being (Gascon et al., 2017; Vian et al., 2021). Evidence shows that residents living near waterfronts tend to report better overall health, lower psychological stress, and greater life satisfaction and happiness (Wheeler et al., 2012, 2015; White et al., 2013), alongside more opportunities for physical exercise and social participation (Pasanen et al., 2019; Ashbullby et al., 2013). Building upon existing models of the health benefits of green spaces, White et al. proposed that blue spaces can improve individual health and well-being through multiple pathways, including reducing environmental stress, strengthening connections with nature, and facilitating attention restoration (White et al., 2020). Despite this growing recognition, research on the health and well-being benefits of waterfronts remains comparatively

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limited relative to the extensive scholarship on urban green spaces. This gap hampers a fuller understanding of the associations between waterfront environments and human health and constrains their incorporation into health-oriented planning and policy practices (Grellier et al., 2017; Hunter et al., 2023). As a result, elucidating how waterfront spaces relate to residents' well-being and how to improve their environmental quality has become a key interdisciplinary concern across environmental psychology, public health, and urban planning.

Subjective well-being (SWB) is described as a psychological state that encompasses individuals' evaluative judgments about their lives and their affective responses to personal experiences (OECD, 2013). A deeper understanding of SWB holds fundamental significance, as it provides a scientific basis for formulating and evaluating broad social policies. Recent research consistently shows that waterfront spaces can substantially enhance individual well-being: more frequent visits, longer dwell times, and higher-intensity participation are all associated with higher levels of SWB (Vitale et al., 2022; White et al., 2021). Traditionally, these studies have relied on self-reported questionnaires and interviews that are costly and time-consuming, thereby limiting scalability for population-level applications (McDougall et al., 2022; Vitale et al., 2022; White et al., 2021). Conceptually, SWB is typically framed as comprising two interrelated dimensions: (i) cognitive evaluations of one's life as a whole, commonly termed life satisfaction; and (ii) experienced well-being, namely the stream of affective states (e.g., pleasure, tension) arising in everyday contexts (National Research Council, 2013). Studies increasingly use the intensity and frequency of momentary affective states as indicators of experienced well-being, assuming that affective responses directly reflect and quantify individual happiness in specific contexts (Y. Lin et al., 2022; Tu et al., 2026). Building on this perspective, the present study operationalizes SWB using sentiment-derived emotional expressions from social media data, focusing on its affective dimension.

It is noteworthy that the health-related effects of waterfront environments vary considerably depending on their types, scales, and functions. Existing studies have predominantly compared waterfronts based on waterbody types (Wang et al., 2024). For instance, McDougall et al. (2022) found that frequent visits to rivers and canals are associated with better mental health, whereas lake visits show no significant association. Luo et al. (2023) further revealed that during viewing and walking activities, rivers and lakes generally offer stronger restorative effects than canals. However, such classifications centered solely on waterbody type fail to capture the complexity and diversity of contemporary urban waterfronts in terms of their spatial morphology, functional attributes, and patterns of use. To address this limitation, recent research has begun to adopt more spatially grounded approaches, examining specific categories of waterfront spaces. For example, Zhang et al. (2024) examined Guangzhou's Pearl Riverfront, using street view segmentation, POI data, and heatmaps to quantify visual indices and explore drivers of waterfront vitality. Hu et al. (2026) focused on Shanghai's Huangpu River Greenway, applying seasonal wearable street view imagery and random forest models to reveal how built environments and seasonal microclimates shape use intensity. Li et al. (2025) investigated Shanghai's waterfront parks, using social media text mining and sentiment analysis to identify key cultural ecosystem services and differentiated drivers of user satisfaction. These studies, focusing respectively on riverfront, waterfront greenway, and waterfront park, reflect a broader academic shift from “waterbody-based classification” to “space-type classification,” offering new perspectives for understanding how different types of waterfront spaces are used and how they influence health outcomes.

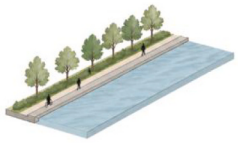
The internal quality and evaluation of waterfront spaces are crucial for a comprehensive understanding of their relationships with health and well-being (Bell et al., 2021). Case studies from Plymouth in the United Kingdom and Barcelona in Spain have demonstrated that improvements in spatial quality can significantly enhance users' happiness and life satisfaction (Van Den Bogerd et al., 2021; Vert et al., 2019). In recent years, some scholars have begun to quantify the quality of waterfront spaces from an objective environmental perspective. For instance, Chen and Yuan (2020) quantified the fragmentation of blue spaces within a one-kilometer buffer and combined field surveys with street-view images to assess hydrophilic characteristics and analyze their impacts on human health. Garrett et al. (2023) proposed an evaluation framework encompassing indicators such as safety, cleanliness, facility integrity, and the presence of wildlife. Similarly, Cao et al. (2024) incorporated both visual elements—such as sky, water, vegetation, and buildings—and spatial factors—such as land use, building density, and the accessibility of transportation and service facilities—to examine their effects on public emotions.

In recent years, the proliferation of social media data has provided new opportunities to examine the interactions between humans and urban spaces (S. Chen et al., 2022a; Shen and Li, 2023). Social media platforms offer large-scale, wide-coverage, and cost-effective data sources that can capture individuals' natural expressions in real-world contexts (Y. Lin et al., 2022; X. Zhu et al., 2021). Meanwhile, advances in natural language processing (NLP) and machine learning have further enhanced the precision and interpretability of text mining, enabling researchers to extract experiential features, identify emotional tendencies, and uncover the complex relationships between them (Li et al., 2024). These developments provide a novel methodological approach to address the representativeness and timeliness limitations of traditional approaches and to explore the complex associations between waterfront environments and public happiness.

Despite increasing research attention, the relationship between urban waterfront environments and SWB remains insufficiently understood. First, existing studies have provided limited insights into the differences among various types of waterfront spaces. Previous research has commonly classified blue spaces based on waterbody attributes (e.g., rivers, lakes, coasts, and canals). Although this classification approach effectively captures the hydrological characteristics of waterfront environments, it may overlook the differences among urban waterfront spaces arising from variations in spatial morphology, functional positioning, and patterns of use. For example, waterfront greenways and waterfront parks may both be located along rivers, yet their distinct spatial organizations and functional characteristics can generate different behavioral patterns and experiential processes (Vian et al., 2021). How these spatial and functional differences are reflected in public experiences and further associated with SWB remains insufficiently explored. Therefore, this study classifies urban waterfront spaces into three typical types based on spatial forms and planning functions: linear waterfront greenways, belt-shaped riverfront spaces, and areal waterfront parks (Y. Chen et al., 2022b), and systematically compares the differences in experiential characteristics and SWB associations across these waterfront typologies (Table 1).

Second, existing studies on the health benefits of waterfront spaces have predominantly adopted an objective environmental perspective, with limited attention given to the experiential processes through which individuals interact with environments. Previous research has commonly used objective indicators, including environmental exposure, landscape characteristics, and spatial accessibility, to characterize waterfront environments and further examine their associations with health outcomes. These studies have provided valuable insights into identifying health-promoting environmental attributes; however, they often provide limited explanations of how individuals perceive, interpret, and interact with spatial environments. Environmental psychology theories suggest that the effects of environments on human health and well-being are not achieved through direct pathways alone, but rather through processes involving environmental perception, cognitive appraisal, and behavioral responses (Leslie et al., 2010; Luo et al., 2021). Compared with objective environmental attributes, individuals' subjective perceptions of environments are often more direct predictors of spatial use decisions and behavioral choices (Leslie et al., 2010), while behavioral activities also serve as an important pathway linking

Table 1
Data for 14 waterfront study sites.

Type	Site	Reviews (n)	Period	Spatial morphology
Waterfront greenway (8643)	a. East Lake Greenway	8643	2019/01/14–2024/12/30	 Linear
Riverfront (4936)	b. Hankou Jiangtan	1553	2019/02/06–2024/12/30	 Belt
	c. Wuchang Jiangtan	1314	2019/01/12–2024/12/24	
	d. Qingshan Jiangtan	1185	2019/01/26–2024/12/23	
	e. Hanyang Jiangtan	884	2019/01/03–2024/12/26	
Waterfront park (6841)	f. Sha Lake Park	2356	2019/01/05–2024/12/27	 Areal
	g. Ziyang Lake Park	1263	2019/02/08–2024/12/27	
	h. Moshui Lake Park	603	2019/01/05–2024/12/29	
	i. Moon Lake Park	556	2019/12/03–2024/12/26	
	j. Lingjiao Lake Park	469	2019/01/11–2024/12/26	
	k. Huanzi Lake Park	447	2019/01/23–2024/12/26	
	l. Northwest Lake Park	391	2019/01/05–2024/12/27	
	m. Tangxun Lake Park	383	2019/01/27–2024/12/30	
	n. Simeitang Park	373	2019/01/22–2024/12/30	

Note: “Jiangtan” refers to Wuhan's characteristic riverfront embankment spaces that integrate flood control, recreation, and urban landscape functions.

environmental conditions with health outcomes (Liu et al., 2019; Murrin et al., 2023). Therefore, reliance solely on objective environmental indicators may not sufficiently explain why different waterfront spaces generate differentiated well-being experiences, nor reveal how experiential factors are associated with SWB. This highlights the need to move beyond objective environmental assessments and examine how multidimensional experiences, including perceptual, cognitive, and behavioral dimensions, are associated with SWB (Y. Cao et al., 2025; Ogawa et al., 2024).

Finally, existing studies still lack a systematic analytical framework capable of identifying waterfront experiences from large-scale user-generated content and further revealing their associations with SWB. Although the integration of social media data and deep learning techniques has opened new analytical pathways for related research, current studies have mainly focused on applying general NLP models for sentiment analysis (Cheng et al., 2021; Huai et al., 2023; Wei et al., 2024) or environmental satisfaction assessment (Huai et al., 2023; Li et al., 2024; Tang et al., 2023), while the identification of specific public environmental experiences and the processes through which they relate to well-being outcomes remain relatively limited. Some studies have further employed keyword matching and dictionary-based approaches to extract environmental experience-related factors. Nevertheless, such methods rely heavily on predefined rules and may fail to adequately capture complex semantic relationships, contextual information, and multidimensional experiential expressions embedded in natural language (I. Kong et al., 2023; Zheng et al., 2024). Therefore, leveraging advanced deep learning approaches to automatically identify public perceptual, cognitive, and behavioral experiences from large-scale textual data and uncover their associations with SWB remains an important research challenge.

To address the above research gaps, this study takes typical waterfront spaces in Wuhan, China, as the research objects and employs social media review data combined with NLP and machine learning techniques. From a public perspective, the study conducts a fine-grained identification of individuals' perceptual, cognitive, and behavioral characteristics in waterfront environments and further explores how these features are associated with SWB in waterfront environments. By comprehensively analyzing both overall and typology-specific variations in SWB and experiential characteristics, this study provides new theoretical insights and empirical evidence for the optimization of waterfront design and health-oriented urban planning. It also seeks to address three key research questions:

(1) How can deep learning models be utilized to achieve a fine-grained identification of public experiences in waterfront spaces?
(2) How is SWB in waterfront environments associated with multi-dimensional features?
(3) What significant differences exist in the experiential association patterns and seasonal rhythms across different waterfront typologies in relation to SWB?

2. Data collection and preprocessing

2.1. Study area

Wuhan, the capital of Hubei Province and a major national hub city, had an urban population of approximately 13.6 million in 2022 (Yuan et al., 2025). Situated at the confluence of the Yangtze River and its largest tributary, the Han River, and dotted with 166 lakes, the city is known as the “City of a Hundred Lakes” (Yongli et al., 2014). This hydrological configuration, “two rivers converging, lakes encircling”, has forged a prototypical waterfront urban form. Prior studies have documented close links between Wuhan's waterfront spaces, residents' health, and social interaction, providing a solid empirical basis for the present case study (J. Wu et al., 2019; Xie et al., 2022). Given its extensive blue-space resources, distinctive hydromorphological configuration, and strong representativeness in waterfront research, Wuhan serves as an exemplary case city for this study. The Third Ring Road of Wuhan broadly delineates the boundary between the urban core and the suburban areas. Within the Third Ring, the city exhibits higher development intensity and a greater concentration of urban functions and public facilities, whereas areas beyond the ring are relatively less developed (Y. Chen et al., 2022b). Accordingly, this study confines its scope to the waterfront spaces located within the Third Ring Road (Fig. 1).

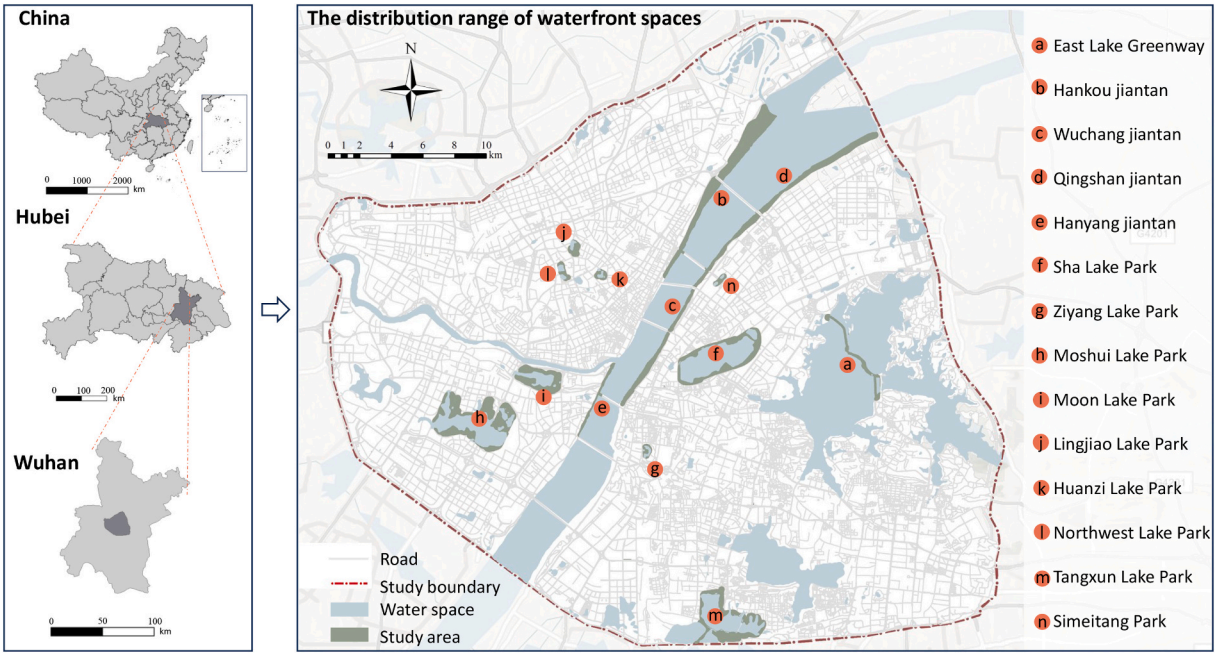


Fig. 1. Study area. Basemap and administrative boundaries: Tianditu Administrative Division Visualization Service (Map approval number: GS (2024) 0650). Road-network data: OpenStreetMap contributors under the Open Database License (ODbL).

2.2. Data sources and processing

As one of China's leading local lifestyle platforms, Dianping aggregates a large volume of user-generated content that captures users' subjective evaluations and experiences of urban spaces, providing an important data foundation for this study (Yuan et al., 2025). This study focuses on urban waterfront spaces in Wuhan, China, and collects user review data from the Dianping platform. Initially, 33,760 reviews from 35 waterfront spaces were obtained, covering the period from 2006 to 2024. Considering user privacy protection requirements, this study only used publicly available and anonymized information relevant to spatial experience analysis, including place labels, publication dates, and review texts. No personally identifiable information was collected or analyzed. To improve data quality and sample representativeness, a multi-stage quality control framework was implemented, including spatial sample selection, temporal filtering, text preprocessing, and noise elimination. First, regarding spatial sample selection, only waterfront spaces with at least 400 raw reviews (prior to data cleaning) were retained to ensure sufficient data availability for subsequent analysis. This threshold helped exclude locations with limited visitor activity and insufficient user feedback, thereby reducing the instability of text-based analyses caused by small sample bias. After this screening process, 14 waterfront spaces were selected as study sites (Fig. 1). These sites cover major waterfront areas along the Yangtze River and Han River, as well as urban lakes of different scales, providing a representative sample of the diverse spatial forms and functional characteristics of Wuhan's waterfront environments. Second, temporal filtering was conducted. Because the number of reviews before 2019 was relatively limited, and data accumulation patterns and platform usage behaviors varied substantially across earlier years, only reviews published between 2019 and 2024 were retained for subsequent analysis. This period provided a more consistent temporal basis for examining seasonal variations and temporal dynamics in waterfront experiences. Finally, systematic text cleaning and preprocessing were conducted. Specifically, duplicate reviews and texts lacking meaningful semantic information were removed, including content consisting solely of emojis, punctuation marks, numerical strings, corrupted characters, and advertisements. Chinese text segmentation was then performed using the Jieba tokenizer. A customized stop-word list was further developed based on word-frequency statistics to filter out auxiliary words, informal online expressions, and symbolic information with limited semantic contribution. Through this multi-stage quality control process, a final dataset of 20,420 valid reviews was obtained, providing a reliable data foundation for subsequent analysis (Table 1; see Appendix A for data preprocessing details).

3. Method

To systematically characterize public experiences in waterfront spaces and further investigate the associations between experiential factors and SWB, this study develops an integrated analytical framework combining deep learning and explainable machine learning (Fig. 2). The framework consists of five major steps. First, based on social media review data, text preprocessing and optimization of semantic embedding models were conducted to construct a high-quality textual corpus and generate semantic representations capturing contextual information embedded in reviews. Second, topic modeling and semantic clustering approaches were employed to identify public waterfront experience themes from large-scale review texts. These themes were further refined through semantic calibration and manual validation to construct a structured experiential indicator system, enabling the transformation of unstructured textual information into multidimensional experiential indicators. Third, an attention-weighted method based on semantic similarity was developed to quantitatively evaluate the semantic associations between review texts and experiential indicators. Fourth, sentiment analysis models were applied to identify emotional expressions in review texts and extract emotion labels and sentiment scores as indicators of experienced well-being. Finally, explainable machine learning models were employed to examine the associations between experiential factors and SWB, identifying key contributing factors and their interaction effects. Through this analytical framework, this study establishes a complete workflow from public experience identification, indicator construction, and emotional quantification to the interpretation of SWB associations, providing systematic empirical evidence for understanding the relationship between urban waterfront environments and public well-being.

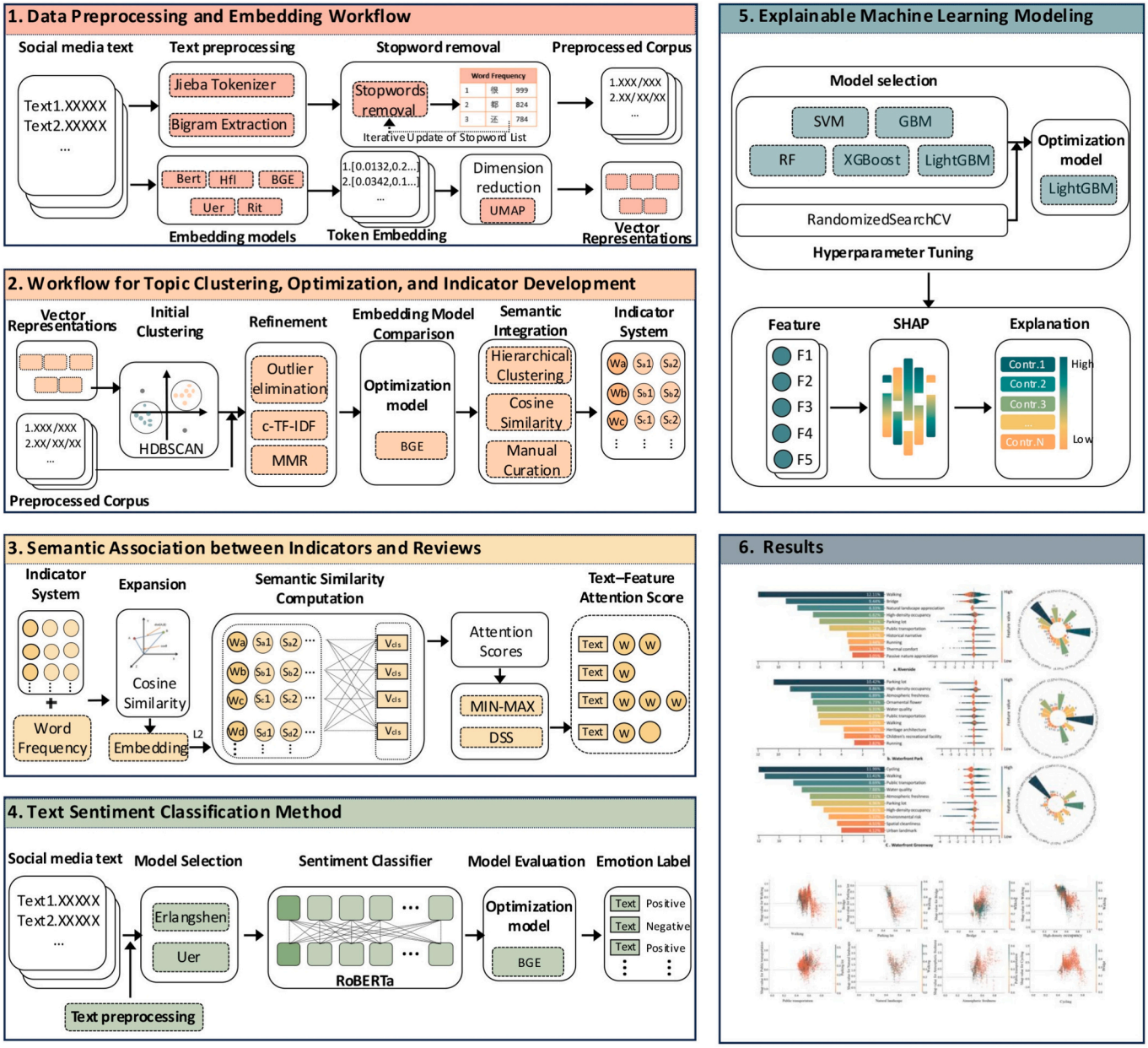


Fig. 2. Research framework.

3.1. Data preprocessing and embedding process

BERTopic is a topic modeling framework built upon Bidirectional Encoder Representations from Transformers (BERT), integrating deep semantic embeddings with topic extraction techniques to adaptively identify distinct semantic categories and extract representative keywords (Raman et al., 2024). To ensure high-quality topic discovery and semantic differentiation, this study performed targeted module optimization and parameter tuning within the BERTopic framework, tailored to the characteristics of waterfront space research and the specific analytical requirements of this study.

We systematically preprocessed the raw review texts to ensure the quality and consistency of the corpus. The jieba tool was used for Chinese word segmentation, and a bigram model was introduced to link high-frequency word pairs, enhancing semantic coherence. Subsequently, considering the significant impact of segmentation quality on topic modeling and clustering results, we optimized the stopword list through word frequency analysis to remove irrelevant or redundant terms. After several rounds of filtering, the resulting corpus preserved the core semantics of waterfront spaces while effectively reducing noise, providing a reliable input for subsequent modeling.

The quality of text embeddings directly influences the effectiveness of topic clustering. To comprehensively capture the deep semantic features contained in the review texts, five embedding models were selected, compared, and evaluated based on their semantic expressiveness, adaptability to Chinese corpora, and embedding stability. Subsequently, Uniform Manifold Approximation and Projection (UMAP) was applied for dimensionality reduction, aiming to preserve the global semantic structure while improving clustering and visualization performance. The resulting low-dimensional semantic space not only facilitates the intuitive visualization of semantic distributions but also establishes a solid foundation for subsequent topic modeling and indicator system construction (see Appendix B for details).

3.2. Topic clustering and indicator system construction

The HDBSCAN algorithm was applied for density-based adaptive topic clustering, and the class-based TF-IDF (c-TF-IDF) method was subsequently used to enhance the representativeness and discriminative power of topic keywords. In the clustering process, some texts exhibited weak semantic coherence and could not be assigned to any topic; these were treated as outliers. To improve the stability and interpretability of the clustering results, a cosine similarity–based outlier filtering mechanism with adaptive thresholding was implemented, which effectively reduced noise interference and clarified topic boundaries. Additionally, the Maximal Marginal Relevance (MMR) algorithm was applied to rank topic keywords, maintaining a balance between representativeness and diversity while preventing dominance by high-frequency terms. Parameter tuning and performance comparisons were conducted across different embedding models using UMAP visualization, through which BAAI/bge-large-zh-v1.5 was identified as the optimal embedding model for semantic clustering. This model subsequently served as the foundation for semantic calibration and indicator system construction (see Appendix C for technical details and clustering results).

Although unsupervised clustering can extract semantic categories from the public perspective, its interpretability and accuracy remain limited. Therefore, this study proposed a multi-level semantic calibration strategy. First, cosine similarity between word embeddings was applied to merge semantically similar keywords, thereby improving topic coherence and compactness (Eq. 1). Second, the clustering results were manually reviewed and semantically refined. To minimize subjective bias, a multi-coder cross-validation procedure was adopted. Researchers (KW and DX) independently organized and integrated the identified topics, while researchers (XZ and HY) conducted an independent review and refined topics with ambiguous boundaries or semantic overlap, thereby improving the clarity and interpretability of the final topic classification. Finally, the semantic similarity between topic keywords and their corresponding seed words was calculated, and the average value was defined as the semantic aggregation index, providing a quantitative measure of topic consistency (Eq. 2).

$$\mathrm{CosSim}(w_i, s) = \frac{\vec{v}_i \cdot \vec{v}_s}{\left\|\vec{v}_i\right\| \cdot \left\|\vec{v}_s\right\|} \tag{1}$$

$$Cohesion(T) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{CosSim}(w_i, s) \tag{2}$$

where w_i and s denote a keyword and its corresponding seed word. $\vec{v}_i$ and $\vec{v}_s$ represent their corresponding word embedding. T represents a given topic, and n indicates the number of seed words contained in that topic.

Through the integrated application of semantic fusion, keyword filtering, and semantic aggregation analysis, this study constructed a comprehensive evaluation indicator system from the public perspective. The framework covers three overarching dimensions: Environmental Perception, Environmental Cognition, and Behavioral Activity, comprising 21 subdimensions and 63 indicators in total (Table 2). Methodologically, the system fully leverages the fine-grained feature extraction capability of BERT-based models, enabling a more comprehensive and accurate representation of the public's multifaceted experiences in waterfront spaces. Moreover, it provides a solid methodological foundation for subsequent analyses (see Appendix D for detailed definitions and descriptions of each indicator).

Table 2
Structured indicator framework.

Dimension	Subdimension	Indicator keywords
a. Environmental perception	1. Vegetation Landscape Perception	Tree (0.76), Grassland (0.71), Ornamental flower (0.75)
	2. Animal Diversity Perception	Bird (0.76), Aquatic animal (0.70), Insect (0.76), Pet (0.70)
	3. Architectural Landscape Perception	Pavilion (0.75), High-rise building (0.80), Bridge (0.74), Corridor (0.72), Platform (0.73), Stair (0.78)
	4. Environmental Cleanliness Perception	Spatial cleanliness (0.81), Water quality (0.72)
	5. Public Service Facilities Perception	Seating facility (0.74), Sanitation facility (0.80), Recreational facility (0.80), Sports facility (0.78)
	6. Traffic Perception	Parking lot (0.80), Vehicle (0.78), Ferry (0.74), Public transportation (0.75)
	7. Auditory Perception	Natural sound (0.65), Anthropogenic sound (0.67)
	8. Olfactory Perception	Atmospheric freshness (0.76), Odor (0.80)
	9. Thermal Perception	Thermal comfort (0.81), Heat (0.80), Cold (0.76)
	10. Crowding Perception	High crowd density (0.79), Low crowd density (0.76)
b. Environmental cognition	11. Safety Cognition	Flood risk (0.70), Environmental risk (0.75), Night illumination (0.69)
	12. Supporting Services Cognition	Catering service (0.72), Commercial service (0.75), Accommodation service (0.74)
	13. Spatial Structure Cognition	Spatial morphology (0.80), Spatial configuration (0.76)
	14. Spatial Design Cognition	Waterside-oriented design (0.74), Creative spatial design (0.72), Stylistic characteristic (0.75)
	15. Spatial Renewal Cognition	Construction and renovation (0.80)
	16. Spatial Image Cognition	Urban landmark (0.77), Site-specific attraction (0.74)
	17. Cultural and Emotional Cognition	Historical narrative (0.70), Heritage architecture (0.84), Emotional memory (0.72)
c. Behavioral activity	18. Leisure and Relaxation Activity	Passive nature appreciation (0.78), Camping and outdoor dining (0.75), Recreational engagement (0.83), Photographic engagement (0.77)
	19. Physical Activity	Dance (0.75), Cycling (0.79), Running (0.78), Ball game (0.76), Walking (0.79)
	20. Social Interaction Activity	Parent–child interaction (0.76), Couple interaction (0.71), Group cohesion activity (0.78)
	21. Observational Experience Activity	Natural landscape appreciation (0.71), Cultural landscape appreciation (0.79)

3.3. Evaluation of semantic associations between the indicator system and review texts

To quantitatively assess the semantic relationships between the constructed indicator system and reviews, this study proposes an attention-weighted method based on semantic similarity. The approach integrates semantic expansion, vector representation, and dynamic filtering to achieve high-precision alignment between indicators and text semantics. The procedure consists of four steps.

(1) Lexicon Expansion. For each indicator's seed-word set, semantically similar words were identified from a high-frequency vocabulary based on cosine similarity (Eq. 1), thereby expanding lexical coverage and enhancing the indicator's ability to capture relevant textual content. This process resulted in a comprehensive seed-word set for each indicator.
(2) Semantic Vector Representation. Review texts and keywords were encoded using the BAAI/bge-large-zh-v1.5 embedding model. Mean pooling and L2 normalization were subsequently

applied to project all embeddings into a unified semantic space, ensuring consistency in vector similarity calculations.

(3) Attention Score Calculation. For each indicator, we calculated the cosine similarity between all seed-word embeddings $W_i = \{s_{i1}, s_{i2}, \ldots, s_{im}\}$ and the mean-pooled embedding of the review text v_{CLS}. Specifically, each seed word within an indicator was compared with the review embedding to generate a set of similarity scores reflecting the semantic relevance between the review and the indicator-related concepts. The mean of the three highest similarity values was defined as the attention score $AttentionScore_i$ (Eq. 3), capturing strong semantic correlations while avoiding dilution from low-similarity words. This process produced a set of semantic relevance scores between each review and all indicators, denoted as AttentionScores (Eq. 4). Consequently, each review can be transformed into a multidimensional indicator vector consisting of the AttentionScores of all indicators, where each AttentionScore quantifies the semantic association strength between the review and a given indicator and is treated as a continuous variable in downstream analyses.

(4) Dynamic Selection Strategy. To avoid overmatching in short texts and information dilution in long texts, this study introduced a dynamic filtering strategy based on min–max normalization (Eq. 5), which adaptively controlled the number of retained indicators according to text length. This strategy ensured that each review was represented only by the most representative indicators, thereby improving the precision and robustness of semantic matching (see Appendix E for details).

$$AttentionScore_i = \frac{1}{k}\sum_{j\in \mathrm{Top}-k}\left(e_{ji}\cdot v_{\mathrm{CLS}}\right) \tag{3}$$

$$\mathrm{AttentionScores} = \{\mathrm{AttentionScore}_1, \ldots, \mathrm{AttentionScore}_n\} \tag{4}$$

$$x_{\mathrm{new}} = \frac{x - \min(x)}{\max(x) - \min(x)} \tag{5}$$

3.4. Measurement of subjective well-being

In this study, SWB is operationalized based on sentiment-derived emotional expressions from social media data, capturing its affective dimension. RoBERTa is an improved variant of the BERT pre-trained language model that enhances training performance through a larger corpus, extended training duration, and optimized training strategies. It has demonstrated superior accuracy and generalization capabilities across various NLP tasks, particularly in sentiment classification (Briskilal and Subalalitha, 2022). To ensure accuracy and applicability in Chinese sentiment analysis, this study compared two open-source RoBERTa-based models pre-trained on large-scale Chinese corpora: IDEA-CCNL/Erlangshen-RoBERTa-110 M-Sentiment (Erlangshen) and UER/roberta-base-finetuned-dianping-chinese (UER). A total of 1500 review samples were randomly selected for manual annotation to evaluate the models' classification performance on positive and negative emotions. Five standard evaluation metrics were employed: Accuracy, Precision, Recall, F1-score, and AUC (Area Under the ROC Curve). The evaluation results are shown in Table 3. Although the UER model exhibits a slightly lower Recall compared with the Erlangshen model, it outperforms Erlangshen on the key evaluation metrics of Accuracy, Precision, F1-score, and AUC, demonstrating superior overall performance and greater stability. Therefore, the UER model was adopted as the primary sentiment classification tool in this study. Based on its output, standardized emotion labels and emotion scores were generated for subsequent quantitative analyses. The emotion labels were treated as binary categorical variables (see Appendix F for details).

In addition, following previous studies (Gao and Fang, 2025; Tu et al., 2026), we introduce the Happiness Intensity Index (HII) as an operational proxy for the affective dimension of SWB (Eq. 6). The index measures the overall intensity of happiness for a given space by calculating the proportion of positive reviews to total reviews within a specified time interval *T*, thereby reflecting the general happiness level in waterfront environments. HII ranges from 0 to 1 and represents a continuous measure of happiness intensity. In this context, "happiness" refers specifically to the affective component of SWB captured by HII. Where N_{positive} and N denote the numbers of positive reviews and total reviews, respectively.

$$HII_{At} = \frac{N_{positive}}{N} \tag{6}$$

3.5. Explainable machine learning modeling

After completing text feature extraction and sentiment classification, the dataset was partitioned into training, validation, and test sets in a 64:16:20 ratio using stratified sampling. To ensure representative and stable model selection, five typical machine learning algorithms were compared in terms of performance: Support Vector Machine (SVM), Gradient Boosting (GB), Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Light Gradient Boosting Machine (LightGBM). RandomizedSearchCV was employed to perform randomized hyperparameter search within predefined parameter spaces, evaluating 15–25 hyperparameter combinations for each model. A three-fold stratified cross-validation procedure was used to further optimize model performance and generalization ability. To mitigate the effect of class imbalance on performance evaluation, this study adopted macro-averaging to compute five metrics: Accuracy, Precision, Recall, F1-score, and AUC. As shown in Table 4, LightGBM achieved the best overall performance across multiple metrics, demonstrating both strong discriminative power and high stability. Therefore, LightGBM was selected as the core model for subsequent feature importance analysis (see Appendix G for details).

To further interpret the model's internal mechanisms and enhance result transparency (T. Hu et al., 2023), this study introduces the Shapley Additive exPlanations (SHAP) method, which improves the model's interpretability from a game-theoretic perspective. In this study, each input feature is considered a participant in the game, with the goal of collectively determining the prediction output. Shapley values (Eq. 7) are used to measure each feature's fair contribution to the prediction, reflecting the marginal contribution of the feature to the prediction for an individual sample. Additionally, Shapley interaction values (Eq. 8) further decompose the joint contribution of feature pairs, revealing the interaction relationships between different features (I. Kong et al., 2023).

$$\Phi_i = \sum_{S\subseteq N\setminus\{i\}} \frac{|S|!(M-|S|-1)!}{M!}\left[f_x(S\cup\{i\}) - f_x(S)\right] \tag{7}$$

Table 3
Sentiment model evaluation results.

Model	Accuracy	Precision	Recall	F1-score	AUC
Erlangshen	0.9067	0.8867	0.9236	0.9048	0.9073
UER	0.9333	0.9379	0.9222	0.9300	0.9329

Table 4
Machine learning model evaluation results.

Model	Accuracy	Precision	Recall	F1-score	AUC
SVM	0.87	0.68	0.63	0.65	0.71
Gradient Boosting	0.88	0.71	0.64	0.67	0.78
Random Forest	0.87	0.69	0.68	0.69	0.78
XGBoost	0.87	0.70	0.68	0.69	0.77
LightGBM	0.88	0.71	0.68	0.69	0.78

$$\Phi_{ij}(f,x) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(M-|S|-2)!}{2(M-1)!} \nabla_{ij}(S) \quad (8)$$

where φ_i and φ_{ij} denote the Shapley value and Shapley interaction value, respectively; M is the total number of features, N is the feature set, and S is a subset of N excluding feature i.

4. Results

4.1. Characterizing HII patterns of urban waterfronts

4.1.1. Overall HII patterns

When all data across different waterfront types were aggregated, the overall HII reached 0.882, indicating that the public generally experiences a high level of SWB in waterfront spaces. Weekly aggregation provided finer insights into temporal dynamics. As shown in Fig. 3a, the median HII was 0.90, with an interquartile range (IQR) of 0.85–0.95 and a minimum value of 0.70, suggesting that HII was consistently concentrated within a high range, with only occasional declines. In terms of temporal trends (Fig. 3b), the HII increased steadily during 2019–2020, maintaining a level of around 0.95, then declined markedly between 2020 and 2022, reaching its lowest point (0.80) in mid-2021. Thereafter, it gradually recovered and stabilized above 0.90 from 2023 onward. Short-term fluctuations exhibited a clear seasonal pattern: HII tended to rise slightly or reach short-term peaks in spring (March–May). It then declined markedly in summer (June–August) to a seasonal trough, rebounded in autumn (September–November), and remained relatively high in winter (December–February) despite a slight decrease.

4.1.2. HII patterns across waterfront typologies

Fig. 4 illustrates the overall distribution and temporal variation of HII across different waterfront typologies. Riverfront exhibited the highest median HII (0.92) with a relatively concentrated distribution (0.89–0.95) and a minimum value of 0.80, indicating that public happiness in these spaces generally remained high, with only minor fluctuations and occasional declines. Waterfront parks showed a slightly lower median (0.91) but the most concentrated distribution (0.89–0.94) and a higher minimum value (0.86), suggesting a marginally lower overall level than riverfront areas but greater temporal stability. In contrast, waterfront greenways had the lowest median (0.90) and considerably lower first quartile (0.82) and minimum values (0.70), reflecting weaker and more volatile happiness experiences overall (Fig. 4a). From a temporal perspective, both riverfront and park spaces exhibited relatively stable long-term trends, maintaining consistent HII levels around 0.90–0.93 with limited short-term fluctuations and a generally alternating upward pattern. In comparison, greenways demonstrate the most pronounced volatility, characterized by recurring phases of decline followed by recovery (Fig. 4b). In terms of seasonal patterns, riverfronts displayed distinct seasonal fluctuations: HII peaked in late spring to early summer (May), dropped to a trough (0.87) in late summer (July–August), gradually rebounded during autumn, and declined slightly again in winter (December–January). Parks remained the most stable, maintaining values between 0.90 and 0.93 across most months, with minimal seasonal variation. Greenways, however, exhibited a typical "U-shaped" seasonal pattern—relatively high at the beginning of the year (January–February, around 0.90), declining steadily to a summer low (August, occasionally as low as 0.75–0.80), and then gradually recovering to around 0.90 in autumn and winter (Fig. 4c).

4.2. Relative feature importance analysis based on SHAP

4.2.1. Overall feature importance

Fig. 5 illustrates the feature importance results across multiple subdimensions derived from the LightGBM–SHAP framework, along with the ranked contributions of the top 25 features and their corresponding SHAP interpretations. All importance scores are computed using normalized mean absolute SHAP values. Collectively, these top 25 features account for 89.11% of the total normalized mean absolute SHAP importance, indicating that the key experiential features have been effectively and comprehensively captured. Overall, the Physical activity subdimension exhibited the highest relative importance (19.12%). Among these behavioral indicators, Walking emerged as the most influential feature, also ranking as the single most important variable across all predictors (11.73%), followed by Cycling (5.06%) and Running (2.33%). The Traffic Perception dimension ranked second, with Parking lot (7.90%) and Public transportation (6.45%) both contributing more than 5% to overall importance. Crowding perception ranked third, where both High crowd density (6.58%) and Low crowd density (3.32%) showed notable contributions. Other subdimensions, including Architectural landscape perception (7.35%), Environmental cleanliness perception (5.97%), Observational experience activity (5.76%), and Olfactory perception (5.13%), also exceeded 5% in relative importance. In contrast, Auditory perception (0%), Animal diversity perception (0.03%), and Spatial renewal cognition (0.51%) showed minimal relevance, each contributing less than 1% to the overall model.

Further SHAP dependence plots revealed the association patterns between individual indicators and SWB. Among all features, Walking not only showed the highest contribution but also had a strong positive contribution to model outputs. Similar positive associations were observed for Public transportation, Atmospheric freshness, Natural landscape appreciation, and Bridge. In contrast, several indicators showed negative associations, including High crowd density, Parking lot, and Environmental risk. Notably, features such as Cycling and Water quality exhibited context-dependent effects, with their association with SWB shifting between positive and negative depending on situational conditions.

4.2.2. Waterfront typologies feature importance

Fig. 6 and Fig. 7 illuminate the relative importance of features in

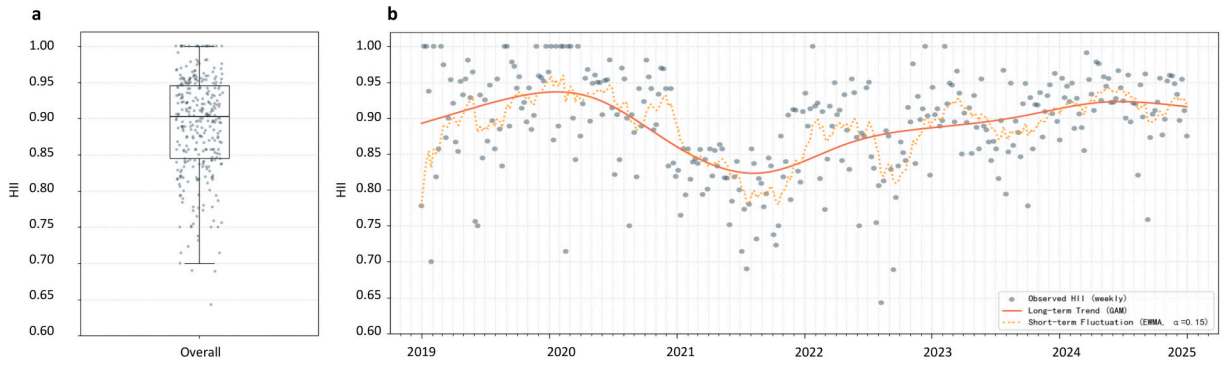


Fig. 3. (a) Overall HII distribution; (b) Temporal variation trend.

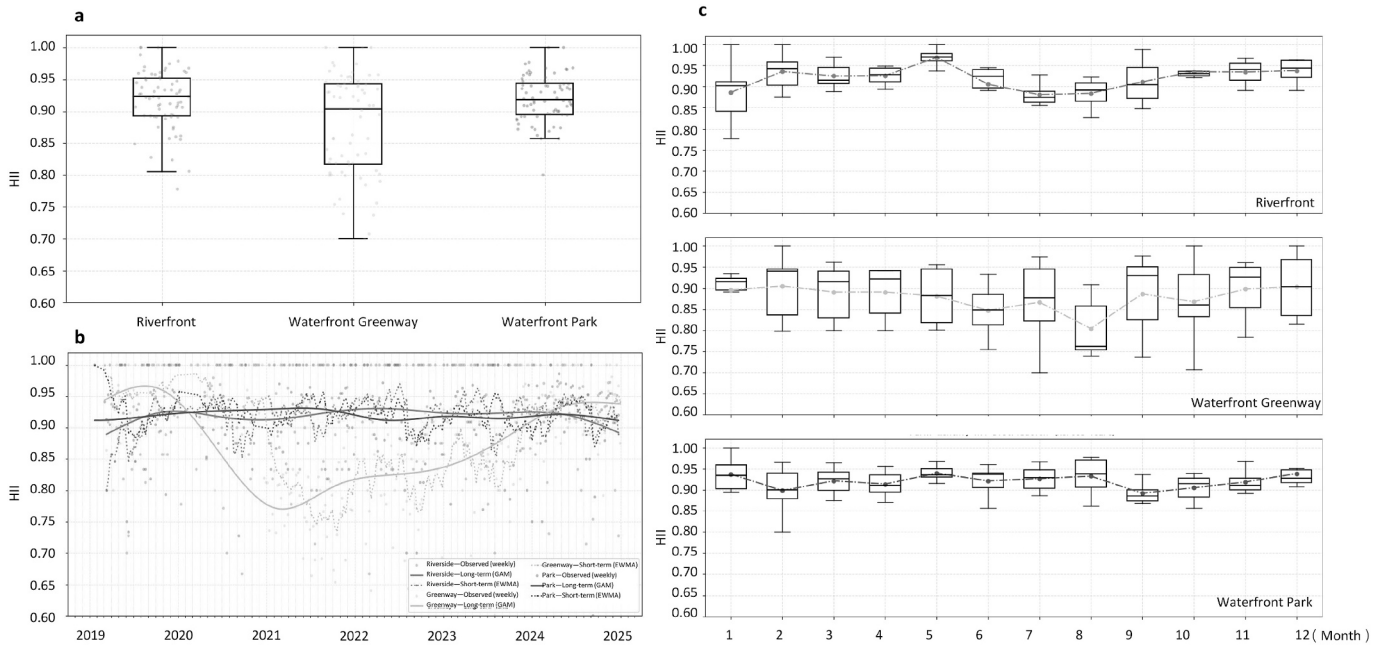


Fig. 4. (a) Overall HII distribution; (b) Long-term and short-term temporal trends; (c) Monthly variation patterns.

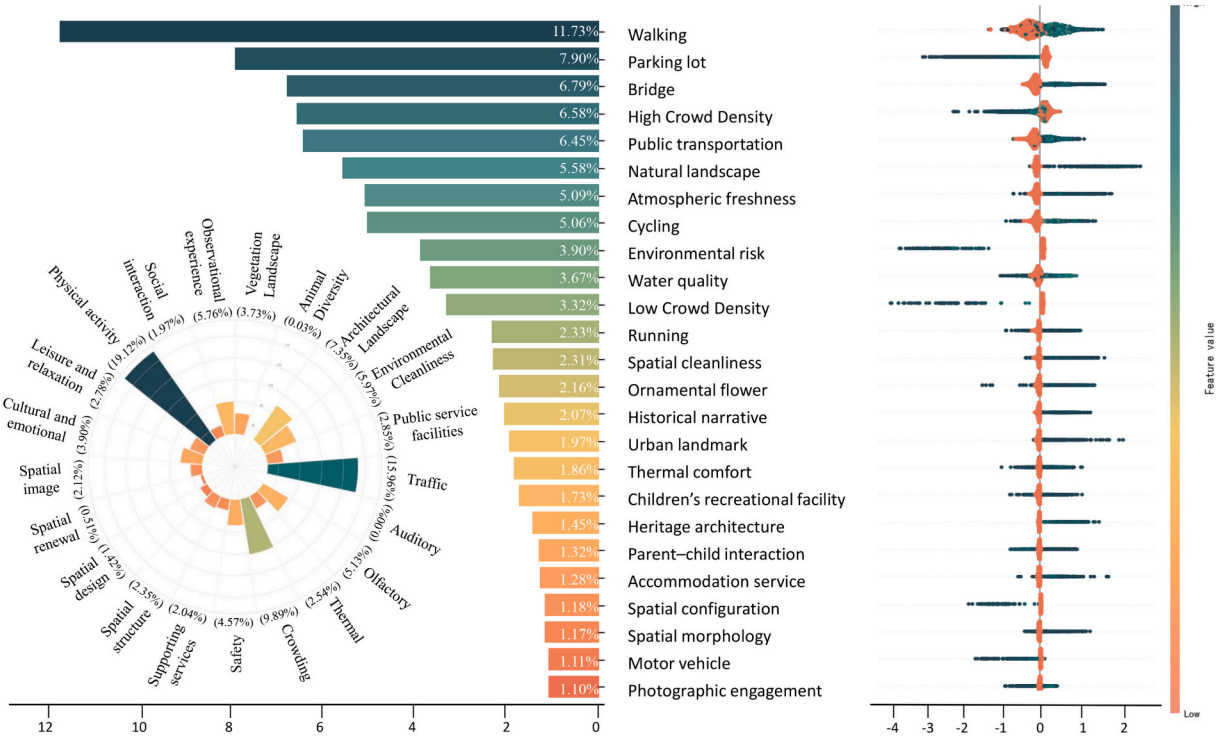


Fig. 5. SHAP-based feature importance for the overall sample.

waterfront settings at two analytical scales: the former compares commonalities and differences across riverfront, waterfront park, and waterfront greenway typologies, while the latter further details site-level patterns within each typology. Overall, Physical activity (e.g., Walking), Traffic Perception (e.g., Parking lot, Public transportation), and Crowding perception (e.g., High crowd density) consistently rank among the most influential features across typologies. Walking and Public transportation generally contribute positively to SWB, while Parking lot tends to contribute negatively. By contrast, High crowd density is more context-dependent: it is predominantly negative in greenways and parks, but on riverfronts can, under certain social or festive conditions, translate into a positive contribution to SWB.

Within the set of waterfront typologies examined, the SWB of riverfronts is strongly associated with Physical activity (15.90%), with Walking (12.11%) as the core contributor and Running (3.44%) also adding a positive effect. Beyond activity, Architectural landscape perception is salient across riverfronts, with Bridge exhibiting particularly high importance. Meanwhile, Observational experience activity (8.52%) and Leisure and relaxation activity (7.77%) also show strong positive associations. In addition, Cultural and emotional cognition,

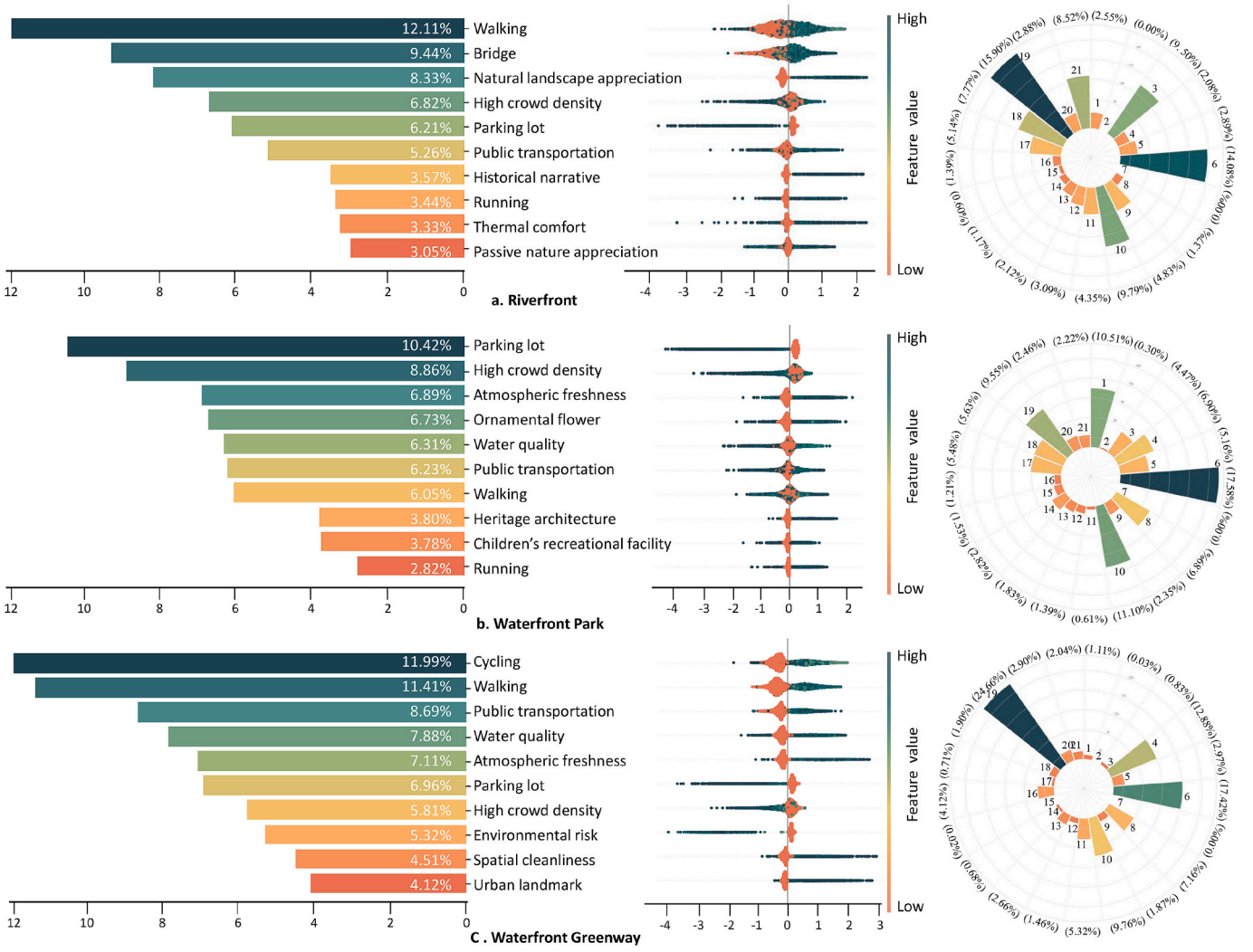


Fig. 6. SHAP-based feature importance across waterfront typologies. Note: The numbers in the radial charts correspond to the indicator IDs listed in Table 2.

such as historical narratives and emotional memory associated with the riverfront, holds considerable relative importance (5.14%). In waterfront parks, transportation perception (17.58%) was the factor most strongly associated with SWB, with Parking lot (10.42%) exhibiting the highest importance among all indicators, highlighting the critical role of accessibility in shaping park experiences. Perceived crowding ranked second (11.10%), among which High crowd density (8.86%) showed particularly high importance, suggesting that visitor activity intensity plays a significant role in well-being experiences. Meanwhile, natural environmental factors also contributed substantially to SWB, with Vegetation landscape perception (10.51%) emerging as one of the most influential indicators. In addition, Olfactory perception (6.59%) and Cultural and emotional cognition (5.48%) also demonstrated notable importance, indicating that well-being experiences in waterfront parks are shaped not only by environmental quality and accessibility, but also by natural perceptions and cultural experiences. The formation of SWB in waterfront greenways is strongly associated with physical activities (24.66%), whose relative importance is significantly higher than that of the other two types of spaces. Among these, Cycling (11.99%) and Walking (11.41%) are the dominant features, with cycling ranking first across all variables, which highlights the core function of greenways in providing high-quality spaces for physical exercise. Beyond common transportation and crowding related features, Environmental cleanliness perception also plays a notable role in greenways, where Water quality (7.88%) and Spatial cleanliness (4.51%) show strong associations. Similarly, olfactory perception, such as Atmospheric freshness (7.11%), shows a strong positive contribution. However, unlike the other two waterfront types, greenways also exhibit distinct negative features, particularly Safety cognition (e.g., Environmental risk), which is strongly associated with lower SWB.

Site-level analyses further revealed variations in the importance of key experiential factors within the same waterfront typology. Among the riverfronts, Wuchang Jiangtan was characterized by natural landscape appreciation, bridge imagery, and walking experiences, forming a composite experience centered on riverscape appreciation and recreational activities. Hanyang Jiangtan was distinguished by its prominent bridge landscapes, together with tranquil natural settings and cultural memory experiences, reflecting strong landscape and cultural characteristics. In contrast, Hankou Jiangtan exhibited stronger statistical associations with social ambiance and group interactive activities, with its experiential profile dominated by family-focused recreation and social leisure. Qingshan Jiangtan combined physical activity with landscape experiences, where walking, bridge scenery, landmarks, and photography-related activities jointly shaped its distinctive experiential profile. Waterfront parks exhibited more diverse well-being-related characteristics. Sha Lake Park, Northwest Lake Park, and Ziyang Lake Park were primarily associated with vegetation landscapes, flower appreciation, and family-oriented activities, highlighting their functions as spaces for nature appreciation and social interaction. Simeitang Park, Huanzi Lake Park, and Moon Lake Park integrated historic architecture,

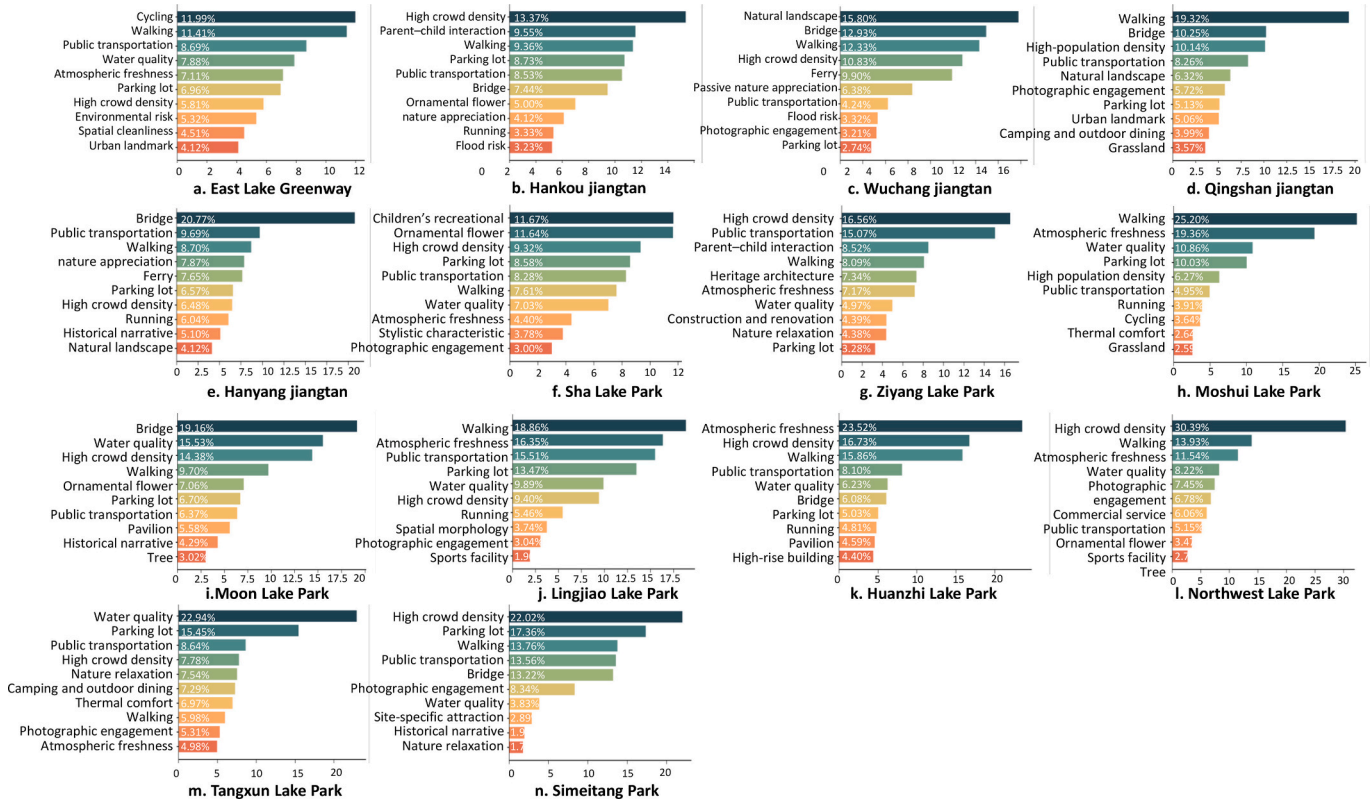


Fig. 7. SHAP-based feature importance at the site level.

cultural narratives, and landscape imagery, demonstrating strong cultural and historical value. Lingjiao Lake Park and Moshui Lake Park were characterized by jogging, cycling, and sports facilities, forming multifunctional spaces that combine physical exercise with natural landscape experiences. Tangxun Lake Park presented prominent associations with camping, dining and thermal comfort perceptions, corresponding to its suitability for long-duration visits and seasonal leisure activities.

4.2.3. *Nonlinear effects and feature interactions*

The associations between different features and SWB are often not purely linear but exhibit complex nonlinear variations and disparities. Fig. 8 illustrates the response patterns of the top eight features, showing how their associations with SWB vary across different intensity levels. The results reveal substantial differences in the relationships between feature attention and SHAP values. Features such as Bridge and High crowd density generally exhibit a linear relationship, where the association with happiness consistently increases or decreases with greater attention, showing distinct and stable trends. In contrast, Walking, Parking Lot, Cycling, and Natural Landscape exhibited pronounced nonlinear response patterns. Specifically, Walking reached its maximum positive contribution at an attention level of approximately 0.58, after which its positive effect gradually diminished, indicating a clear pattern of diminishing marginal returns. Parking Lot showed a distinct turning point around 0.57, beyond which its negative influence tended to stabilize. Cycling, Natural Landscape, and Atmospheric Freshness exhibited

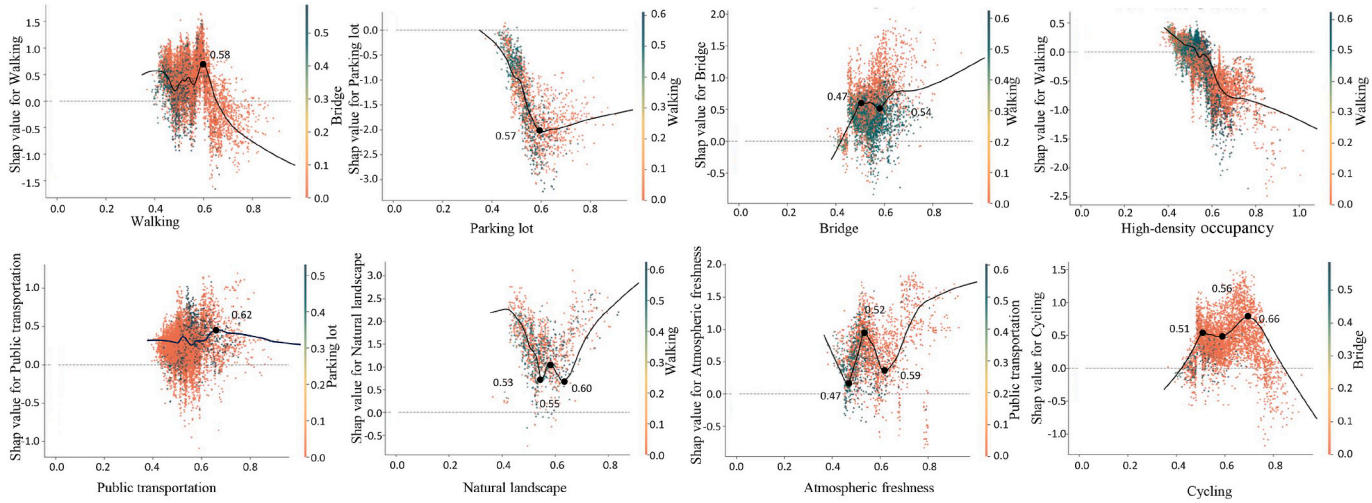


Fig. 8. Relationships and interaction patterns of key features.

multiple critical change points, suggesting stage-dependent effects on SWB. These findings indicate that the contributions of these experiential factors to well-being do not increase indefinitely with higher attention levels, but are instead constrained by contextual conditions and experiential thresholds.

In addition, a single text often encompasses multiple features whose complex interactions are associated with its emotional outcome. Fig. 8 further illustrates the interaction patterns among the top eight core features and their strongest interacting counterparts. The results indicate several positive synergistic relationships. For instance, the presence of Bridge in waterfront spaces enhances the positive contribution of Walking to SWB. Meanwhile, negative interactions are also observed; for example, strong Public transportation tends to weaken the positive contribution of Atmospheric freshness to SWB. These findings suggest that different combinations of environmental and behavioral factors are associated with differentiated emotional outcomes, highlighting the context-dependent nature of waterfront experiences.

To further explore potential heterogeneity across different waterfront typologies, grouped SHAP dependence analyses were conducted for riverfront, waterfront parks, and waterfront greenways. The results generally supported the findings from the full-sample analysis, with the response trends of key indicators remaining largely consistent across typologies, suggesting the robustness of the associations between major experiential factors and SWB. Nevertheless, differences were observed in the primary interaction variables of several indicators across waterfront types. For example, walking primarily interacted with bridge perception in riverfronts, whereas it showed a stronger interaction with High crowd density in waterfront greenways. These findings suggest that although the overall association patterns of core experiential factors remain relatively stable, their specific interaction patterns are context-dependent and influenced by the spatial characteristics and functional attributes of different waterfront environments (see Fig. S2 in Appendix H).

5. Discussion

5.1. HII patterns and feature importance in waterfront spaces

From a temporal perspective, variations in waterfront HII are jointly associated with social events and climatic conditions. The long-term trend shows that the decline from 2020 to 2022 was likely associated with reduced spatial use caused by mobility restrictions during the COVID-19 pandemic (X. Wu et al., 2023), whereas the rapid rebound after 2023 reflects the recovery of the happiness effect of waterfront spaces following the restoration of social order. This rebound vividly demonstrates the role of waterfronts as vital public environments that facilitated the restoration of residents' SWB and everyday normality during post-crisis recovery, representing an important manifestation of urban resilience (Grace et al., 2024). Short-term fluctuations exhibit a seasonal pattern: the waterfront happiness index tends to decline under extremely high or low temperatures, which to some extent suggests a correlation between thermal conditions and the waterfront happiness index. Research has shown that weather factors such as sunlight, temperature, and precipitation can indirectly influence happiness by affecting individuals' physiological states, emotional experiences, and social interactions (X. Wu et al., 2023). Despite these temporal fluctuations, the overall HII of waterfront spaces has remained consistently high over the long term, highlighting their enduring role in emotional regulation and psychological support within urban environments.

Physical activity was identified as the experiential factor most strongly associated with SWB in waterfront spaces, and this key finding is consistent with the prevailing conclusions of existing research. Previous studies have generally demonstrated that waterfront environments provide residents with opportunities for walking, running, cycling, and other forms of physical activity, while higher levels of activity participation are typically associated with better health outcomes and higher SWB levels (Bell et al., 2021; Murrin et al., 2023). However, previous research has rarely systematically compared the relative contributions of physical activity with other multidimensional experiential factors, including perceptual, cognitive, and behavioral dimensions. This study further reveals that, among all identified experiential factors, physical activity exhibits the strongest association with SWB, highlighting behavioral experience as a key experiential mechanism underlying the association between waterfront environments and well-being. This finding suggests that behavior is not merely an external manifestation of individual states but also constitutes an essential component in the formation of positive well-being experiences (Y. Liu et al., 2019; Murrin et al., 2023). From the perspective of Person–Environment Fit Theory, waterfront spaces typically provide long, continuous activity corridors and open visual environments, which better satisfy residents' needs for exercise and recreational activities (Caplan and Van Harrison, 1993; French, 1974). Such a high degree of compatibility between environmental resources and individual needs facilitates physical activity engagement and generates comfortable and pleasurable experiences, representing an important advantage of waterfront spaces compared with other urban open spaces (Benton et al., 2021). This person–environment fit not only promotes participation in physical activities but also enhances individuals' positive experiences during activities, thereby contributing to higher levels of SWB. Meanwhile, behaviors related to relaxation, social interaction, and aesthetic appreciation also showed substantial contributions, confirming the multifunctional and composite health value of waterfront spaces beyond their role as simple exercise venues. This finding further expands current understandings of the diverse pathways through which blue spaces support human well-being.

Moreover, beyond behavioral patterns, perceptual features of the built and social environments are also strongly associated with people's happiness experiences in waterfront spaces. Among these factors, Traffic Perception and Crowding Perception exhibited the greatest contributions. Previous studies have primarily focused on objective accessibility indicators, such as road-network density and spatial distance, to examine the positive effects of accessibility on waterfront use frequency and health benefits (Tieges et al., 2022; Zhan et al., 2026). In contrast, this study uses Traffic Perception to capture individuals' experiences of transportation convenience and accessibility. This finding extends existing accessibility research by incorporating the human-centered experiential dimension and provides a more comprehensive understanding of how accessibility contributes to well-being benefits in waterfront environments. Regarding Crowding Perception, existing studies have mainly focused on external built-environment characteristics and internal spatial quality, while the impacts of population density and subjective crowding experiences on well-being remain relatively underexplored (H. Cao et al., 2024; Gao and Fang, 2025; Garrett et al., 2023; G. Zhu et al., 2023). This study further reveals a bidirectional and nonlinear relationship between perceived crowding and SWB, suggesting that the effects of crowding are highly context-dependent. This phenomenon can be interpreted through the lens of Environmental Load Theory. Individuals are continuously exposed to environmental stimuli, and their psychological responses depend largely on the intensity of such stimulation. Moderate levels of environmental stimulation can enhance spatial vitality and social engagement, thereby generating positive emotional experiences, whereas excessive stimulation may exceed individuals' coping capacity and induce stress, fatigue, and cognitive overload (Milgram, 1970; Stokols, 1972). Therefore, high population density does not necessarily result in negative experiences. In contexts such as festivals, public events, and social gatherings, moderate crowd concentration may enhance place vitality and strengthen the social atmosphere, thereby promoting positive emotions. However, excessive crowding may increase psychological burden and reduce SWB. This provides a possible explanation for the bidirectional association between crowding and SWB observed in this study. Notably, this study found that soundscape had limited independent explanatory

power for SWB, which differs from previous studies highlighting the important role of soundscape characteristics in shaping environmental experiences and well-being (F. Liu et al., 2022; J. Zhang et al., 2025). This discrepancy may be attributed to the representational characteristics of social media text data. Compared with visual landscapes and activity-related experiences, soundscape experiences are more transient, context-dependent, and less explicitly expressed in textual narratives. Consequently, soundscape-related experiences may be underrepresented in user-generated content.

Further analysis revealed substantial interaction patterns among waterfront features in relation to SWB. Some features exhibited positive synergistic associations—for example, a positive interaction between Bridge and Walking was observed in their contribution to SWB. Bridges not only increase the visual appeal of the environment but also provide extended pathways and visual focal points, thereby enriching the walking experience (Plasencia-Lozano, 2015). In contrast, public transportation showed a negative interaction with Atmospheric freshness in their contribution to SWB, likely because frequent traffic flow and associated emissions negatively affect people's subjective perception of air quality. These findings indicate that SWB is associated with multiple features rather than a single factor, and their relationships vary under specific contextual conditions.

5.2. SWB differences and association patterns across waterfront typologies

Different types of waterfront spaces exhibited substantial variations in both HII levels and temporal dynamics, which were closely associated with differences in their experiential characteristics. Previous studies have typically classified waterfront spaces according to waterbody attributes, such as rivers, lakes, and canals, and compared the associations between different waterbody types and health-related outcomes (Gao and Fang, 2025; F. Liu et al., 2022; Luo et al., 2023; McDougall et al., 2022). This study extends existing research by advancing both the classification paradigm and theoretical interpretation of waterfront–health relationships. Specifically, it not only identifies variations in well-being levels and seasonal dynamics across different waterfront space types but also further explores the potential explanations for these differences.

Specifically, riverfronts demonstrate the highest overall HII level, accompanied by moderate seasonal fluctuations, which are associated with a diversified behavioral structure—particularly the combination of physical activities and recreational leisure—linked to positive emotions and psychological restoration. Broad and continuous waterfront trails, together with unobstructed spatial layouts, are strongly associated with physical activity and higher levels of positive emotions and mental vitality (Yan et al., 2024). Meanwhile, leisure activities such as enjoying the river breeze or appreciating the scenery create typical scenes of emotional restoration and sensory pleasure, allowing visitors to relax and alleviate mental fatigue and cognitive load through visual and auditory stimulation (Severin et al., 2022; G. Zhu et al., 2023). The architectural and landscape features of riverfronts, particularly the iconic Yangtze River Bridge, together with the expansive river views and the surrounding riverfront buildings, form a richly layered visual interface. This landscape configuration not only enriches the exercise experience by enhancing the visual appeal of the space but also enhances the waterfront's recreational appeal by reinforcing the city's image and local identity (Lynch, 2023).

The overall HII level of waterfront parks is slightly lower than that of riverfronts, yet it remains the most stable throughout the year, showing the weakest sensitivity to seasonal and climatic variations. Transport accessibility and environmental quality show strong combined associations with SWB in waterfront parks. Compared with riverfronts and greenways, waterfront parks are typically embedded within residential areas on a micro-scale and spatially dispersed pattern, making them the most frequently accessed waterfront environments for daily visits. Therefore, Traffic Perception is the most strongly associated factor with SWB in waterfront parks. The higher the level of transport accessibility, the more frequently and intensively residents visit and use these spaces, thereby significantly enhancing their happiness experiences (*L.* Kong et al., 2022). In addition, vegetation landscapes are a defining feature of waterfront parks. Floral displays and lush greenery are associated with richer sensory experiences and higher perceived environmental quality, which may contribute to psychological restoration (Zhou et al., 2023). Such environments also serve as popular venues for social interactions, including parent–child activities and neighborhood gatherings (Bao et al., 2022; Veitch et al., 2022).

In contrast, waterfront greenways exhibit the lowest overall level of HII, the greatest annual fluctuation, and the strongest dependence on seasonal variation. SWB in greenways is most strongly associated with health-oriented physical activity, which shows the highest relative importance among the three typologies. Cycling and walking showed the highest relative importance among all activity-related indicators, reflecting the strong activity-oriented characteristics of waterfront greenways (He et al., 2021). Their linear layout and strong connectivity create ideal conditions for sustained physical activity, enabling residents to gain both physical vitality and psychological relaxation while staying close to nature (Xie et al., 2021). Moreover, favorable environmental conditions—such as good water quality, cleanliness, and fresh air—act as enabling factors for physical activity, contributing to greater comfort and enjoyment during exercise, as well as higher levels of happiness during such experiences. However, greenways also present notable negative factors, particularly the sense of environmental danger, which significantly suppresses SWB.

From a theoretical perspective, the observed differences in SWB across waterfront space types can be interpreted as the combined effects of Person–Environment Fit, restorative, and environmental load mechanisms. First, different waterfront space types offer heterogeneous behavioral opportunities, which in turn generate distinct activity structures. These behavioral differences can be explained through the Person–Environment Fit framework, emphasizing the match between individual needs and environmental affordances. Furthermore, variations in activity structures are likely to shape how well-being responses in different waterfront environments vary under changing climatic conditions. Previous studies have indicated that, compared with leisure and social activities, physical activity is generally more sensitive to climatic conditions (Ferguson et al., 2023; Turrisi et al., 2021). Accordingly, waterfront greenways, where walking and cycling dominate, exhibit the most pronounced seasonal variation. In contrast, waterfront parks, characterized by more static recreational use and higher vegetation coverage, demonstrate lower climate sensitivity and more stable levels of SWB. Secondly, from the perspective of Attention Restoration Theory, riverfronts, with their open views and diverse activity opportunities, are more likely to evoke key restorative environment characteristics, including being away, fascination, extent, and compatibility (Kaplan and Kaplan, 1989). These qualities in turn facilitate positive emotions and psychological restoration, leading to higher levels of SWB. Finally, from the perspective of Environmental Load Theory, this phenomenon provides a useful framework for understanding how spatial crowding and environmental risks shape SWB. Waterfront parks, due to their relatively limited spatial scale, are more susceptible to environmental load induced by crowding, which partially explains why their overall well-being levels are slightly lower than those of riverfronts (M. Lin et al., 2023; J. Zhang et al., 2023). In addition, waterfront greenways are often characterized by narrow linear configurations, with certain sections experiencing mixed pedestrian–vehicle flows, insufficient lighting, and limited visibility. These environmental conditions are more likely to generate feelings of insecurity and perceived environmental risk (Kwiatkowski and Karbowinski, 2023). Elevated perceptions of environmental risk may increase individual vigilance and psychological stress, thereby weakening positive emotional experiences and restorative effects. This further explains why

greenways exhibit lower overall SWB compared with riverfronts and waterfront parks. This finding further extends the analytical scale of blue-space research.

5.3. Implications for theory and practice

This study broadens the analytical scope regarding the linkage between urban waterfront spaces and SWB, providing novel empirical evidence for understanding heterogeneous health benefits across diverse waterfront environments. Theoretically, this study advances existing waterfront classification approaches by shifting the conventional waterbody-based categorization toward a framework grounded in spatial morphology and functional attributes. It systematically classifies urban waterfront spaces into three representative types: riverfronts, waterfront parks, and waterfront greenways, and further reveals the differences among these spatial types in terms of SWB association patterns and seasonal dynamics. Furthermore, by integrating person–environment fit theory, restorative environment theory, and environmental load theory, this study provides theoretical interpretations of the differentiated relationships between waterfront types and SWB. Beyond previous studies that have mainly focused on objective built environment attributes, this research shifts attention toward human-centered experiential dimensions by developing a refined waterfront experience indicator framework encompassing environmental perception, environmental cognition, and behavioral activities. The framework enables a systematic examination of the associations between multidimensional experiential factors and SWB. The findings suggest that the health benefits of waterfront environments are not solely determined by physical environmental characteristics but are closely associated with how individuals perceive, evaluate, and interact with these environments. This experience-based typological framework enriches existing blue space health research and provides new empirical insights into the differentiated well-being benefits of diverse waterfront spaces.

Practically, the findings provide actionable and health-oriented guidance for typology-specific waterfront planning and policy-making. First, physical activity emerges as the primary driver of waterfront SWB. Spatial design should therefore prioritize continuous and high-quality walking, running, and cycling networks, alongside improvements in lighting, safety, and resting facilities to enhance the health-promoting potential of waterfront environments. Meanwhile, Traffic perception and perceived crowding represent fundamental conditions for positive user experiences. Enhancing transport connectivity, optimizing pedestrian flow, and improving spatial carrying capacity can effectively strengthen accessibility and reduce negative perceptions of overcrowding. Second, differentiated strategies are required across waterfront typologies. Riverfronts should leverage open spatial structures and iconic landscapes to integrate physical activity with scenic appreciation. Waterfront parks should strengthen accessibility, recreational amenities, and cultural narratives to foster socially and emotionally engaging environments. Waterfront greenways, by contrast, should maintain their activity-oriented functions while prioritizing safety and environmental quality—such as improved lighting, monitoring, and water management—to address their higher sensitivity to environmental constraints. Finally, waterfront SWB reflects the combined effects of multiple environmental and behavioral factors. Planning should therefore move beyond single-factor optimization toward integrated and adaptive strategies, balancing key elements and recognizing potential threshold effects. From a policy perspective, both equity and differentiation are essential: ensuring equitable access to basic activity and leisure opportunities while reinforcing the distinctive functions of different waterfront types. This approach supports more efficient resource allocation, enhances public happiness, and promotes healthier and more sustainable urban development.

5.4. Limitations and future work

Although this study provides an innovative perspective and methodological framework for analyzing the relationship between waterfront spaces and SWB, several limitations remain. From a data perspective, first, there is a potential sample selection bias. Active users on the Dianping platform are predominantly young and middle-aged urban residents with relatively high digital literacy, while older adults and minors are underrepresented in online expressions (Ghermandi and Sinclair, 2019; Su and Sun, 2025). As a result, the dataset does not fully capture all age groups and social strata, which may systematically underestimate the health needs and experiential perceptions of vulnerable populations regarding waterfront spaces. Second, there are limitations in data dimensionality. Due to platform privacy regulations and restrictions on data access, this study could not obtain key sociodemographic variables such as age, gender, income level, and educational background. This constraint prevents the conduct of heterogeneity analyses and limits the ability to reveal differentiated patterns of SWB across social groups. Consequently, the effects of certain experiential factors on health outcomes may be misestimated, thereby constraining the external validity and generalizability of the findings (Huang et al., 2025).

From the perspective of measurement, the assessment of SWB in this study is relatively unidimensional. This study primarily relies on text-based sentiment analysis to capture public emotional expressions, focusing on immediate affective experiences, while failing to fully encompass the multidimensional nature of health effects associated with waterfront spaces. In fact, the health benefits of waterfront environments extend beyond momentary emotions to include life satisfaction, psychological and physiological restoration, and social well-being. Sole reliance on textual sentiment data therefore cannot adequately represent the comprehensive health value of waterfront spaces (Diener, 1984).

From a methodological perspective, this study primarily relies on social media text data to examine human–environment interactions, which inevitably imposes certain limitations on both analytical scope and causal inference. While textual data can effectively capture users' subjective experiences and evaluations, they are unable to comprehensively represent objective environmental exposure characteristics of waterfront spaces, such as visual landscape attributes, activity settings, and other physical environmental conditions. Furthermore, the findings of this study should be interpreted as statistical associations rather than causal effects. Although the SHAP-based explainable machine learning framework can effectively identify the relative contribution of experiential factors and reveal complex nonlinear association patterns, the cross-sectional nature of social media data does not allow causal relationships between waterfront experiences and SWB to be established.

Future research could be further advanced in three directions. First, mixed-methods approaches combining field surveys and individual-level tracking studies could be employed to address the limitations of social media data in terms of sample representativeness and individual information. Specifically, targeted investigations involving vulnerable groups, such as older adults and children, should be conducted, while collecting detailed individual characteristics, including sociodemographic attributes, visit frequency, duration of stay, and travel modes, to better reveal heterogeneous waterfront health experiences across different populations. Second, future studies could establish a multidimensional framework for assessing health effects by moving beyond single-dimensional emotional measurements. Psychological scales such as the WHO-5, GHQ-12, and perceived restoration measures, together with wearable health monitoring devices and health records, could be integrated to comprehensively evaluate the psychological, physiological, and social benefits of waterfront environments. Third, future research could incorporate multimodal big-data approaches by integrating image-based analysis to extract environmental characteristics, such as vegetation coverage, sky openness, and visual complexity, with

spatiotemporal behavioral data, including mobile positioning trajectories and stay duration. Such data fusion could provide deeper insights into the mechanisms through which waterfront environments influence public health and well-being, while offering more robust evidence-based support for urban public space planning and policymaking.

6. Conclusion

This study integrates large-scale social media data, deep learning, and explainable machine learning approaches to investigate the association patterns between SWB and public experiences across different types of urban waterfront spaces. By developing a comprehensive analytical framework encompassing perceptual, cognitive, and behavioral dimensions, the study provides a novel perspective for understanding the interactions between human health and urban environments. The results indicate that SWB in waterfront spaces is closely associated with multiple experiential factors, including behavioral, perceptual, and cognitive experiences, among which behavioral activities play a dominant role. Moreover, distinct well-being characteristics and association patterns were observed across different waterfront typologies, suggesting that the relationships between experiential factors and SWB are highly context-dependent. These findings highlight the importance of examining typological differences among waterfront spaces and provide valuable empirical evidence for the optimization of urban waterfront environments and health-oriented design practices.

Declaration of generative AI in scientific writing

Statement: During the preparation of this work the authors used GPT-based language model in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Ke Wang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation, Conceptualization. **Xia Zhang:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Hong Yang:** Writing – review & editing, Supervision, Investigation. **Luliang Tang:** Writing – review & editing, Supervision, Funding acquisition. **Dong Xie:** Writing – review & editing, Validation, Conceptualization. **Cunyu Yuan:** Writing – review & editing, Validation, Data curation.

Ethical approval

This study was approved by the Ethics Committee of Wuhan University (Approval No. WHU-NS-IRB2025001). The research used publicly available social media data from Dianping. All data were anonymized during processing, and no personally identifiable information was collected or analyzed.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ecolind.2026.115367.

Data availability

Data will be made available on request.

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